

Verified fast shaking: repeated whole-trajectory cost

The verified fast setting is **6.11 times faster** in median native stepping time than the resolved reference in this predeclared six-run comparison. Every repetition passes unchanged trajectory and physical budgets against both previously qualified references. Histories repeat bitwise within each setting.

Setting	All three native times (s)	Median (s)	Qualified
reference	18.184, 20.211, 17.719	18.184	True
candidate	3.337, 2.809, 2.975	2.975	True

The scene is 27 spherical rigid bodies inside a six-wall container, shaking at plus/minus 20 m/s with reversals at 0.04 and 0.08 s, simulated for 0.12 s. The synthetic pair friction is 0.4, restitution is zero and gravity is 9.81 m/s². Geometry, masses, full inertia, contact law and prescribed motion are identical. The candidate uses travel fraction 0.015, four primary steps and 10 ms output frames; the reference uses fixed 1.25 microsecond internal steps and 5 ms output frames. Comparisons use common 10 ms times. No material parameter was adjusted.

Full RMS budgets: 5 mm position, 0.05 m/s velocity, 0.1 rad/s spin and 0.01 rad orientation. Every run also passes quaternion norm, finite energy change minus measured wall work and actual internal-update surface containment. Both reference ladders independently pass quarter budgets. Original failed refinement evidence remains unchanged in the earlier archive.

Execution source: `63330336b8d0839d971dcd500e5a7ed977defb5e`. Native stepping includes state recording and excludes process/JSON overhead. Whole-process timings and every repetition are retained in the summary and raw histories. Runs launch sequentially in a predeclared interleaved order. Heavy team experiments were paused during execution; external host load cannot be fully controlled. This is a scoped cost comparison on one environment, not a universal engine ranking, experimentally calibrated material result or Vektor-native gain.

The faster setting is selected by full-trajectory evidence, not by assuming a contact residual proves trajectory accuracy. General random-hull trajectories still reject, and arbitrary many-body elastic angular history is not integrated.

Reproduce from the frozen source with `python -m research.run_shake_performance`; independently audit with `python -m research.audit_shake_performance`. Plans, source snapshots, source and binary hashes, six histories, physical checks, repeated-state checks and all timings are retained in `results/`.